

Ireland Housing Intelligence

An Interactive Analytics Platform for Ireland's Housing & Rental Market

Live Application: <https://ireland-housing-intelligence.streamlit.app>

GitHub Repository: <https://github.com/varadkamtikar/Ireland-Housing-Intelligence>

1. Project Overview

Ireland Housing Intelligence is a full-stack data analytics platform developed by Varad Kamtikar, an MSc Data Science candidate at Technological University Dublin. The project was conceived in response to Ireland's ongoing housing crisis — a challenge that affects renters, buyers, and policymakers alike. Rather than relying on anecdotal reporting, the platform draws exclusively on official government data to provide transparent, evidence-based insight into rental and property markets across every county in Ireland.

The platform is built as a multi-page Streamlit application backed by a cloud PostgreSQL database. It ingests rental data from the Residential Tenancies Board (RTB) — the same authoritative source used by Irish government policymakers — and will incorporate the Central Statistics Office (CSO) property price index and earnings data to power affordability analysis. The result is a single, interactive destination for exploring rent trends, property valuations, and housing affordability across Ireland's 26 counties.

2. Platform Modules

The application is structured as four specialised modules, each serving a distinct analytical purpose:

- **Overview Dashboard:** The central command centre of the platform. Displays high-level KPI cards (total rental records, national average rent, years of data coverage, highest- and lowest-rent locations), an all-time national rent trend chart, county-level location rankings, and a property-type breakdown. Designed to give any visitor an immediate, data-grounded read on the Irish rental market.
- **Rent Trends:** A deep-dive analytical page that allows users to filter RTB rental data by year, property type (house, apartment, etc.), and number of bedrooms. Produces KPI metrics and interactive Plotly charts — including bar charts of the most and least expensive locations — making it easy to identify where rents have risen fastest and which areas remain most affordable for specific household configurations.
- **Property Prices:** This module will integrate the CSO Residential Property Price Index at county level, enabling side-by-side comparison of rental yields versus property purchase prices. Users will be able to explore how property valuations have tracked against rent inflation over time, and identify counties where buying may represent better long-term value than renting.

- **Affordability Index:** A forward-looking analytical module that will combine RTB rental data with CSO earnings statistics to produce rent-to-income ratios for each county. It will include an interactive map of Housing Pressure Zones (HPZs) designated by the Irish government, a risk-scoring system indicating which areas are most financially strained, and choropleth visualisations powered by GeoPandas and Folium.

3. Key Features

- **Multi-Page Navigation:** Four-module Streamlit app using the native `st.navigation` API for seamless, SPA-style routing across Homepage, Overview, Rent Trends, Property Prices, and Affordability pages.
- **Official Government Data:** All rental data sourced directly from the RTB (Residential Tenancies Board) — the statutory body responsible for regulating the Irish private rented sector — covering 2007 to 2024 across all counties, property types, and bedroom configurations.
- **Interactive KPI Dashboard:** Headline metrics surfaced at-a-glance: total rental records, national average rent, locations covered, years of data, and highest/lowest-rent counties.
- **Dynamic Rent Trend Charts:** Plotly-powered line and bar charts showing national rent evolution over time with fully interactive zoom, pan, and hover tooltips.
- **Granular Filtering:** Year, property type, and bedroom-count filters on the Rent Trends page enable precise market segmentation — from studio apartments in Dublin to four-bedroom houses in Connacht.
- **Cloud PostgreSQL Backend:** Rental data is persisted in a Neon-hosted PostgreSQL database with SQLAlchemy ORM, enabling efficient querying and a clean separation between the ETL layer and the presentation layer.
- **Shared Cached Data Layer:** A centralised `src/data.py` module serves all pages from a single cached query, preventing redundant database calls and ensuring consistent data across the application.
- **RTB Data ETL Pipeline:** An automated ingestion script (`fetch_rtb_data.py`) loads and cleans the raw RTB CSV data into a structured PostgreSQL schema ready for analytics.
- **CSO Property Price Integration:** An ETL pipeline (`fetch_cso_data.py`) will connect to the CSO API to pull the Residential Property Price Index at county level, enabling the Property Prices module.
- **Interactive Choropleth Maps:** GeoPandas and Folium-powered county-level maps will visualise rent-to-income ratios and Housing Pressure Zone designations across Ireland in the Affordability module.
- **Rent Forecasting Model:** An XGBoost regression model will be trained on historical RTB data to generate short-term rental price forecasts at county and property-type level.
- **Custom Dark-Mode UI:** A bespoke CSS theme applied to all pages — featuring a deep navy-to-indigo hero section, card-based layouts, and a consistent Inter/Calibri typography system — delivers a polished, professional look beyond the default Streamlit appearance.

4. Data Sources

The platform is built on authoritative, government-published datasets — the same sources referenced by Irish housing policymakers:

Source	Dataset	Coverage
Residential Tenancies Board (RTB)	Average Monthly Rent Report	2007–2024 All 26 counties By property type & bedrooms
Central Statistics Office (CSO)	Residential Property Price Index	County-level, monthly — powering the Property Prices module
Central Statistics Office (CSO)	Earnings & Labour Costs Survey	For rent-to-income affordability calculations in the Affordability module

5. Technology Stack

The project is built entirely in Python with a modern open-source data stack:

Layer	Technology
Frontend & UI	Streamlit (multi-page), custom CSS, Inter/Calibri typography
Interactive Charts	Plotly Express & Plotly Graph Objects — line, bar, histogram, scatter, radar
Database	PostgreSQL (Neon cloud) via SQLAlchemy ORM + psycopg2
ETL & Data Wrangling	Pandas — CSV ingestion, cleaning, and feature preparation
Geospatial Mapping	GeoPandas + Folium + streamlit-folium — county choropleth maps
Machine Learning	XGBoost + Scikit-learn — rental price forecasting model
Configuration	python-dotenv for local development; Streamlit Secrets for deployment
Deployment	Streamlit Community Cloud

6. Data Pipeline

The ETL architecture separates ingestion, storage, and presentation into clearly defined layers:

1. RTB publishes Average Monthly Rent CSV reports → `data/raw/rtb.csv` (raw source data)
2. `fetch_rtbt_data.py` ingests and cleans the RTB CSV → PostgreSQL `rental_data` table
3. CSO API → `fetch_cso_data.py` → PostgreSQL `property_prices` table (Property Prices module)
4. CSO Earnings API → ingestion script → PostgreSQL `earnings` table (Affordability module)
5. `src/data.py` provides a shared `@st.cache_data` query layer consumed by all dashboard pages
6. `app.py` orchestrates multi-page navigation; each page reads from the shared data layer and renders Plotly charts

7. XGBoost training pipeline reads historical rental data → trains forecasting model → serialises model artefact

7. Database Schema

The PostgreSQL schema is designed to support multi-dimensional slicing by county, property type, bedroom count, and time:

Column	Type	Description
location	TEXT	Irish county or region
year	INTEGER	Calendar year of the rental period
property_type	TEXT	House, Apartment, etc.
bedrooms	TEXT	Number of bedrooms (1, 2, 3, 4+)
average_rent	FLOAT	Average monthly rent in EUR

Additional tables for property_prices (CSO RPPI) and earnings (CSO Labour Costs) will be added as those modules are completed.

8. Project Structure

The codebase follows a clean separation of concerns across ingestion, data access, and presentation:

File / Module	Purpose
app.py	Application entry point — configures Streamlit navigation across all pages
home.py	Homepage with hero section, stats cards, and feature overview
pages/0_Overview.py	Overview dashboard — KPIs, national rent trend, location rankings
pages/1_Rent_Trends.py	Rent Trends page — year/type/bedroom filters, cheapest and most expensive areas
pages/2_Property_Prices.py	Property Prices module — CSO RPPI integration, price vs rent yield
pages/3_Affordability.py	Affordability Index — rent-to-income ratios, HPZ map, risk scoring
src/db.py	SQLAlchemy engine setup and connection management
src/data.py	Shared cached data query layer (serves all pages)
src/create_tables.py	Database schema initialisation script
src/fetch_rtb_data.py	ETL: RTB CSV ingestion and PostgreSQL loading
src/fetch_cso_data.py	ETL: CSO property price and earnings API ingestion

9. Skills & Competencies Demonstrated

- **Full-Stack Python Development:** End-to-end application from ETL scripts through cloud database to a multi-page, production-deployed Streamlit frontend.
- **Data Engineering:** Automated ingestion pipelines for official government sources (RTB CSV, CSO API) with structured PostgreSQL persistence and a shared caching layer.
- **Interactive Data Visualisation:** Plotly Express and Graph Objects used throughout — line charts, bar charts, histograms, scatter plots, and radar charts — all with interactive zoom, pan, and hover.
- **Geospatial Analytics:** GeoPandas and Folium integration for county-level choropleth maps of rent-to-income ratios and Housing Pressure Zone designations across Ireland.
- **Applied Machine Learning:** XGBoost regression pipeline for short-term rental price forecasting at county and property-type level, trained on 17+ years of RTB historical data.
- **Cloud Database Integration:** Neon-hosted PostgreSQL with SQLAlchemy ORM, credential management via environment variables and Streamlit Secrets, and clean schema design for time-series analytics.
- **Public Policy Data Literacy:** Working with official government datasets (RTB, CSO) and translating policy concepts (Housing Pressure Zones, rent-to-income ratios) into analytical features.
- **UI/UX Design:** Custom CSS theming — deep navy hero section, card-based layouts, and consistent typography — elevating the Streamlit default into a polished, branded product.

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